

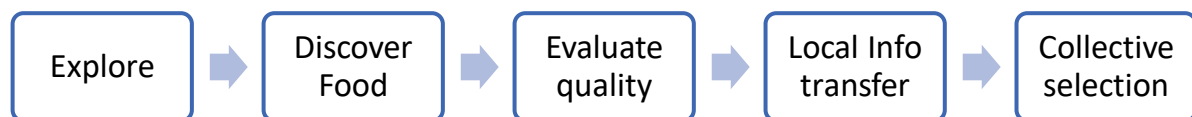
ANT FOOD SOURCE SELECTION

Description of the Biological process:

FORAGING:

Foraging is the process by which animals explore, locate, evaluate and exploit food resources in their environment. In ants, foraging is a decentralized collective behavior. individual ants do not have a complete map of available food sources or a central controller telling them where to go. Instead, ants independently explore their surroundings and respond to locally available information.

During exploration, an ant may encounter a food source and assess its profitability based on factors such as the amount or quality of food and the effort required to reach it. Information obtained by individual ants can influence the behavior of other ants through local interactions. In pheromone-based species, for example, ants can deposit pheromone trails that affect the probability that other ants follow a particular route. As more ants exploit a profitable food source, the associated feedback can strengthen collective preference for that source.



FROM LECTURE:

collect items,

find food source (any target area), exploit it (e.g., establish trail)

subtasks:

exploration, navigation, recruitment, (collective) transport

biological examples: ants, wasps (building material)



IDENTIFICATION OF MECHANISMS TRANSFERABLE TO ROBOTICS:

Below is the direct mapping of Ants behaviour that can be carry overed as robot features.

